

MA.6.NSO.3.1

Benchmark

MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

Example: Middleton Middle School's band has an upcoming winter concert which will have several performances. The bandleader would like to divide the students into concert groups with the same number of flute players, the same number of clarinet players and the same number of violin players in each group. There are a total of 15 students who play the flute, 27 students who play the clarinet and 12 students who play the violin. How many separate groups can be formed?

Example: Adam works out every 8 days and Susan works out every 12 days. If both Adam and Susan work out today, how many days until they work out on the same day again?

Benchmark Clarifications:

Clarification 1: Within this benchmark, expectations include finding the greatest common factor within 1,000 and the least common multiple with factors to 25.

Clarification 2: Instruction includes finding the greatest common factor of the numerator and denominator of a fraction to simplify the fraction.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.3.2, MA.6.NSO.3.4
- MA.6.AR.1.4
- MA.6.AR.3.2, MA.6.AR.3.3, MA.6.AR.3.5

Terms from the K-12 Glossary

- Area Model
- Composite Number
- Dividend
- Divisor
- Factor
- Greatest Common Factor (GCF)
- Least Common Multiple (LCM)
- Prime Factorization
- Prime Number

Vertical Alignment

Previous Benchmarks

- MA.4.AR.3.1

Next Benchmarks

- MA.7.NSO.2.2
- MA.7.AR.1.2

Purpose and Instructional Strategies

In grade 4, students determined factor pairs for whole numbers from 0 to 144 and determined if numbers are prime, composite, or neither. In grade 6, students will determine the least common multiple and greatest common factor of two whole numbers. In future grades, students will use their understanding of factors, factorization and the distributive property to generate equivalent expressions and solve equations.

- Instruction includes a variety of methods and strategies to determine the least common multiple (*MTR.2.1, MTR.3.1, MTR.5.1*).
 - Multiple listing method

6	6, 12, 18, 24, 30
15	15, 30

- Factorization

Use Prime Factorization to find the LCM.

6	$2 \cdot 3$
15	$3 \cdot 5$

The LCM of 6 and 15 is 30 and is represented as $2 \times 3 \times 5$.
 Discuss with students why 3 is listed only once. To use the strategy of prime factorization to find the LCM of two numbers, students must know what to do when a prime factor repeats.

18	$2 \cdot 3 \cdot 3$
60	$2 \cdot 2 \cdot 3 \cdot 5$

The LCM of 180 is represented as $2 \times 2 \times 3 \times 3 \times 5$.

14	$2 \cdot 7$
15	$3 \cdot 5$

The LCM of 210 is represented as $2 \times 3 \times 5 \times 7$.

- Using the Ladder Method (Successive Division)

$$\begin{array}{r|rr}
 2 & 18 & 24 \\
 \hline
 3 & 9 & 12 \\
 \hline
 & 3 & 4
 \end{array}
 = 2 \times 3 \times 3 \times 4 = 72$$

- Instruction includes a variety of methods and strategies to determine the greatest common factor (*MTR.2.1, MTR.3.1, MTR.5.1*).
 - T-charts

96		138	
1	96	1	138
2	48	2	69
3	32	3	46
4	24	6	23
6	16		
8	12		

The greatest common factor of 96 and 138 is 6.

- Factorization

GCF is represented as 3.

6	2×3
15	3×5

Using the Ladder Method for GCF

$$\begin{array}{r}
 2 \overline{) 18 \quad 24} \\
 \underline{3 \quad 9 \quad 12} \\
 3 \quad 4
 \end{array}$$

The GCF of 18 and 24 is $2 \times 3 = 6$. A student may start with any factor that can evenly divide into the given numbers. A student may have started with 3 or 6.

$$\begin{array}{r}
 2 \overline{) 96 \quad 138} \\
 \underline{48 \quad 69}
 \end{array}
 \rightarrow
 \begin{array}{r}
 2 \overline{) 96 \quad 138} \\
 \underline{3 \quad 48 \quad 69} \\
 19 \quad 23
 \end{array}
 \rightarrow
 \begin{array}{r}
 2 \overline{) 96 \quad 138} \\
 \underline{3 \quad 48 \quad 69} \\
 19 \quad 23
 \end{array}$$

The GCF of 96 and 138 is $2 \times 3 = 6$. A student may start with any factor that can evenly divide into the given numbers. Each “rung” of the ladder is created when the reduced numbers have a common factor.

- The greatest common factor can be determined by dividing the numbers multiple times by common factors and then multiplying the common factors that were used as divisors.

$$\begin{array}{r}
 75 \div 5 = 15 \\
 150 \div 5 = 30 \\
 15 \div 3 = 5 \\
 30 \div 3 = 10 \\
 5 \div 5 = 1 \\
 10 \div 5 = 2
 \end{array}$$

The greatest common factor of 75 and 150 can be found by multiplying $5 \times 3 \times 5 = 75$.

$$\frac{75 \div 75}{150 \div 75} = \frac{1}{2}$$

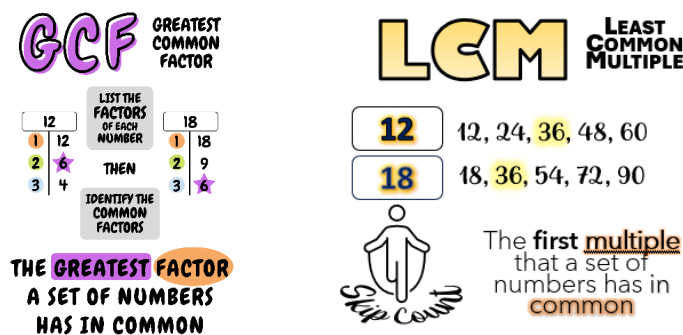
- Instruction includes the use of technology to explore, interpret and define greatest common factors and least common multiple.

Common Misconceptions or Errors

- Greatest common factor and least common multiple are often mixed up or confused by students. Stressing the academic vocabulary of what is a factor versus what is a multiple can help students keep the two straight in their minds.
- Students may incorrectly think that the greatest common factor of two numbers cannot be 1. However, if given two prime numbers, the greatest common factor will always be 1.
- Students may incorrectly think that if one of the numbers is prime, the greatest common factor must be 1. However, the greatest common factor is not 1 if one of the numbers is a prime number and that prime number is a factor of the second number.
- Students may incorrectly think the least common multiple has to be larger than both of the given numbers. Provide examples to showcase the least common multiple being the same as one of the given whole numbers.
 - For example, the LCM of 12 and 24 is 24.
- Students may incorrectly think the least common multiple can be found by multiplying the two numbers. Provide examples where this is not the case, such as the LCM of 4 and 6 is 12 not 24.

Strategies to Support Tiered Instruction

- Teacher shares their thinking when determining if a given situation requires finding the greatest common factor or the least common multiple. Steps include reading the given context out loud and thinking out loud about if the context requires the final answer to be greater than or equal to the given numbers (least common multiple) or less than the given numbers (greatest common factor).
- Teacher creates and posts an anchor chart with visual representations of factors and multiples and encourages students to utilize the anchor chart to assist in utilizing correct academic vocabulary when referring to factors and multiples.
 - Example:



- Teacher provides students with flashcards to practice and reinforce academic vocabulary.

- Teacher provides instruction on the definitions of greatest common factor and least common multiple then co-creates a graphic organizer using student-created definitions, lists of key characteristics, examples, and non-examples of each term.
 - A non-example for the greatest common factor might include finding the least common factor, which would result in 1.
 - A non-example for the least common multiple might include attempting to find the greatest common multiple, which cannot be determined because multiples are infinite.
- Teacher provides examples to showcase the least common multiple being the same as the larger of the given whole numbers or the same as the product of the two numbers.
 - For example, the least common multiple of 12 and 24 is 24.
 - For example, the least common multiple of 14 and 5 is 70.
- Teacher provides examples where this is not the case, such as the least common multiple of 4 and 6 is 12 not 6 or 24.
- Instruction includes the use of a graphic organizer to compare the multiples or factors of two numbers to help recognize common multiples and factors.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.6.1)

Parker reads a book every 12 days and Leah reads a book every 8 days.

Part A. If they both start a new book on Wednesday, how many days will it take for them to start another new book on the same day?

Part B. What day of the week will it be when they both start a new book on the same day?

Instructional Task 2

What is the greatest common factor of 636 and 132? Explain your method for solving.

Instructional Items

Instructional Item 1

What is the greatest common factor of 85 and 187?

Instructional Item 2

What is the least common multiple of 18 and 12?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*